

Lesson 9, Part 2 Activity: Estimate One Slice of Your Project

CS 5388 / CS 4390 Software Project Management | Team activity, complete together

Team: 1 **Project:** Damn Parking

What this activity is for

Your Sprint 2 business case has to answer a question every sponsor asks: how much work would this take, and what would it cost? You are not going to build the system in this course. You are doing what a project manager does before a project is approved: estimating the work so a decision can be made.

Estimating a whole project at once hides mistakes. So you will start small. Pick one or two major parts of your project, called a slice, and estimate only that slice. A small slice lets you count everything, check every number, and see clearly where your assumptions matter.

You will estimate the same slice twice, using two methods that start from different information. Then you will compare the two answers, explain why they differ, and present a range with your assumptions written down. That range becomes the Estimation Appendix on your Sprint 2 portal page.

What to bring

1. Your Release 1 capability list (Step 1 of the Part 1 starter).
2. Your backlog from the Part 1 activity: user stories, MoSCoW categories, and planning poker point sizes.
3. Your similar-products research from Part 1 (small and big products, with timelines and team sizes).
4. The Part 2 slides, especially Slides 13 to 17 (sizing), 19 to 24 (team size), 25 to 27 (story points), and 31 to 33 (reconciling).

How the activity fits together

Step	What you do	What you end up with
1	Pick your slice	One or two major parts of your project, with their stories, screens, and data listed
2	Choose two estimation methods	Two methods that use different starting information
3	Estimate the slice with each method	Two effort estimates, both in team hours
4	Check the estimates for team size	Confidence that coordination costs are included
5	Compare the two estimates	A written explanation of why they differ
6	Present the result	A dated range, in hours, dollars, and weeks, ready for the business case

Key terms

Term	What it means in this activity
Slice	One or two major parts of your project that a user would recognize, such as “buy a permit.” You estimate only this.
Size	How much software the slice contains, measured in function points, use-case points, or story points.
Effort	Total team hours of work. If 5 people each work 10 hours, that is 50 hours of effort.
Productivity rate	Hours of effort per unit of size, such as 8 hours per function point. The rate drives the whole estimate.
Assumption	Any number you chose rather than measured. Label it, so a reader knows it could change.
Velocity	Story points a team completes per sprint. You have no sprint history yet, so yours is an assumption.
Loaded rate	The full hourly cost of one person: salary plus benefits, overhead, and tools.
Range	A low and a high estimate. Honest estimates are ranges, not single numbers.
O, M, P	Optimistic, most likely, and pessimistic estimates, used in the PERT formula $(O + 4M + P) \div 6$.

About the examples

Every italic example uses the same slice from the fictional Campus Parking System: “Buy a semester permit.” Following one slice through every method shows how the pieces connect. Your own numbers come from your project.

Step 1. Pick your slice

A good slice is a part of your project that a user would recognize and that is useful on its own. It usually covers 3 to 8 user stories from your backlog.

How to choose

1. Start with a Must-have capability that is central to your project. If it did not exist, the project would not make sense.
2. If you pick a second part, make it different in kind, such as one part users interact with and one part that produces reports or connects to another system.
3. Do not pick the easiest part. The slice should include some real complexity, such as payments, another system, or stored data.
4. Write down what is out of the slice, so your estimate has a clear boundary.

Question	Our slice
Slice name (one or two major parts)	Live parking monitoring
Why we chose it	We chose this slice because all our features are used to help users with this single goal at its core
Screens or forms where users enter data	Mobile-friendly parking map/list with location and filter controls, selected-lot details with directions, and a manager dashboard.
Reports, results, or lookups users see	List and or virtual map, of available parking spots
Data the slice stores	Parking-lot information, including location, parking type, cost, and capacity; and timestamped occupancy observations/history.
Other systems it connects to	City maps, and existing camera observing parking lots
Out of the slice (not included)	Camera installation, development or training of a new computer-vision model, payments, reservations, and automated enforcement. We assume an existing camera service supplies occupancy observations and that possible misuse is reviewed by a person.

Stories in the slice

User story	MoSCoW	Points
As a driver, I want to view nearby parking lots to quickly find a place to park.	M	6
As a driver, I want to see live parking data to view current availability and compare parking options between different lots.	M	8
As a driver, I want to get directions to my selected parking lot so that I can get there quickly and easily	C	3
As a parking lot manager, I want to observe misuse of parking and parking patterns	M	5
As a driver, I want to filter parking options by distance, parking types, and costs	S	3
As a driver, I want to see the amount of available parking in a given lot, so I don't have to drive to a full lot	M	7

User story	MoSCoW	Points
Total points		32

Example slice, “Buy a semester permit”: a student registers a vehicle (3 points), chooses a lot and permit type (5), pays online (8), receives a digital permit by email (3), and looks up the permit later (2), for 21 points. It stores permit and vehicle records and connects to the student registry and a payment service. Citations and appeals are out of the slice.

Step 2. Choose two methods

Each method starts from different information. Choose two that do not share the same starting point, so that if they agree, the agreement means something (Part 2, Slide 31).

Method	It starts from	A good choice if	Watch out for
Analogous	Similar products and how long they took	You found at least one product close to yours	Copying a big company's timeline without adjusting
Function points	Counts of what users enter, see, and store	Your Step 1 table lists screens, results, and data	A productivity rate with no source
Use-case points	The actors and use cases in your slice	You can list who uses the slice and for what	Default adjustment factors used without saying so
Story points to cost	Your planning poker points and a velocity	Every story in the slice has points	Velocity is a guess until you run sprints

Suggested pairs

1. Function points plus analogous: one starts from your own design, the other from the market. The examples below use this pair.
2. Story points plus analogous: one starts from your backlog, the other from the market.
3. Function points plus story points: one starts from system features, the other from team judgment of story size.

Our two methods:

1. Function Points
2. Story Points

Why do these two use different starting information:

Function points start from counting what users enter, see, and store in the design; on the other hand, story points start from the team's own relative judgment of story size plus an assumed velocity.

Step 3. Estimate the slice

Complete only the two sections (3A to 3D) that match your methods. Whatever method you use, finish with effort in team hours for your slice, so the two estimates can be compared in Step 5. Where you must choose a number, write it down, and label it as an assumption.

3A. Function points (Part 2, Slides 13 to 15)

What it measures: the functionality users get from the slice, counted in five categories. Your Step 1 table already lists most of what you need.

Category	What to count	Example from the permit slice
External Inputs (EI)	Forms where data enters the system	Register vehicle, purchase permit, update vehicle
External Outputs (EO)	Results the system calculates and sends	Permit confirmation and receipt
External Inquiries (EQ)	Simple lookups of stored data	Look up my permit, list lots with permits available
Internal Logical Files (ILF)	Groups of data the slice stores	Permit records, vehicle records
External Interface Files (EIF)	Data read from another system	Student registry, payment service

How to do it

1. Use your Step 1 table. Count each distinct input, output, lookup, and data group in the slice once.
2. Multiply each count by its weight and add the results. That total is your slice size in function points.
3. Choose a productivity rate in hours per function point. You have no history, so label it an assumption and test a low and a high value (Slide 15 shows 6 to 12 hours).
4. Multiply size by each rate to get a low and high effort in hours.

Category	Count	Weight	Points
External Inputs (EI)	1	4	4
External Outputs (EO)	3	5	15
External Inquiries (EQ)	1	4	4
Internal Logical Files (ILF)	2	10	20
External Interface Files (EIF)	1	7	7
Total function points			50

Rate: **low 6 and high 12 hours per FP** Assumption or source: Assumed productivity of 6-12 hours per function point, using the range suggested in the activity. Counts assume an existing camera service provides occupancy data, an external map service provides routing, and the manager dashboard summarizes parking patterns for human review.

Effort: **50 FP × 6 h = 300 hours (low)** **50 FP × 12 h = 600 hours (high)**

Our preliminary counts include one input for occupancy updates, three outputs for calculated availability, manager summaries, and directions, one inquiry for nearby-lot lookup and filtering, two stored data groups for lot information and occupancy history, and one external data group for map/routing information.

3C. Story points to cost (Part 2, Slides 25 to 27)

What it measures: effort based on your team's own sizing of the stories in the slice. Story points are relative, so you convert them using a velocity (points per sprint). Because you have not run any sprints, velocity is an assumption. Use a low, middle, and high value.

How to do it

1. Use the total points from your Step 1 story table.
2. Choose a low, middle, and high velocity in points per two-week sprint. Label them assumptions.
3. Divide the slice points by each velocity. Keep the decimals; a small slice may take less than one sprint.
4. Multiply by team hours per sprint (team size $\times$ hours per person per sprint) to get effort in hours.

Input	Value	Note
Slice points	32	From your Step 1 story table
Team size	4	Five or six at most
Hours per person per sprint	40, assuming 20 hours/week over a 2-week sprint	State your assumption
Velocity: low/middle/high	8 / 10 / 16 per sprint	Assumptions until you have sprint data
Sprints needed (high to low velocity)	$32 / 16 = 2$ $32 / 10 = 3.2$ $32 / 8 = 4$ sprints	Slice points $\div$ velocity
Effort in hours (low to high)	320 / 512 / 640	Sprints $\times$ team size $\times$ hours per person per sprint

Example: 21 points at 30, 20, or 15 points per sprint takes 0.7, 1.05, or 1.4 sprints. With 5 people working 80 hours each per sprint (400 team hours), effort is 280 to 560 hours.

We assume four team members, each available for 20 hours per week. Velocity is assumed because we do not have measured implementation sprint history. All six stories, including Should-have and Could-have stories, are included in the 32-point estimate.

Step 4. Check your estimates for team size (Part 2, Slides 19 to 24)

Larger teams spend more of their time coordinating: meetings, handoffs, and fitting work back together. Estimators tend to underweight this cost, so check each estimate against the list below. Anything marked No needs a fix or a written explanation.

Check	Yes / No	Why it matters
Our team is five or six people at most.	Yes	Beyond six, split into two teams.
We counted our communication paths: $4(4 - 1) \div 2 = 6$ paths	Yes	Five people share 10 paths; ten share 45.
We added explicit work for integrating the pieces of the slice.	Yes	Breaking work into pieces hides the cost of putting it back together.
We did not assume that more people means proportionally more output.	Yes	Coordination grows faster than headcount.
We estimated for the team we have.	Yes	Not for a larger team we wish we had.
We adjusted, not copied, timelines from much larger teams.	Not applicable	Their effort includes their coordination costs.

We assume an additional 40 team hours for integration and coordination: 12 hours for connecting occupancy data, storage, and availability calculations; 12 hours for connecting the interface, map, directions, and dashboard; 8 hours for end-to-end testing and integration fixes; and 8 hours for coordination and handoffs.

Our base estimates cover feature implementation and feature-level testing. This separate allowance is added once to each method.

Function Points: $300\text{--}600 + 40 = 340\text{--}640$ hours. Using an assumed middle productivity rate of 9 hours/FP, the middle estimate is $50 \times 9 + 40 = 490$ hours.

Story Points: $320\text{--}640 + 40 = 360\text{--}680$ hours. The middle estimate is $512 + 40 = 552$ hours.

The larger-team timeline adjustment is not applicable because neither method uses another team's timeline.

Step 5. Compare your two estimates (Part 2, Slides 31 and 32)

Two estimates almost never match. The gap is useful: it points to the assumption you are least sure about. Your job is to find that assumption and name it.

	Method 1	Method 2
Method	Function Points	Story Points
Estimate in hours (low to high)	340 - 640 hours	360 - 680 hours
Middle value in hours	490 hours	552 hours
Input that drives it most	Hours per function point	Points completed per sprint
Is that input measured or assumed?	Assumed	Assumed

Gap: larger middle value ÷ smaller middle value = **1.13** The larger middle estimate is approximately 13% higher.

If one estimate is more than about twice the other, recheck your counts and conversions before explaining the gap.

Questions to answer

1. Do your two methods depend on the same assumption? If so, their agreement proves little.
They use different starting information: Function Points use functional counts, while Story Points use backlog sizes and assumed velocity. However, both share the same scope, team context, and 40-hour allowance. Their agreement is a useful cross-check, not independent proof of accuracy.
2. Which single input moves the answer most? Try changing it and see.
Productivity assumptions drive both methods. Holding the 50 FP count fixed, changing the rate from 6 to 12 hours/FP changes the final effort from 340 to 640 hours. Holding capacity fixed, changing velocity from 16 to 8 points per sprint changes the final effort from 360 to 680 hours. Velocity produces a slightly wider range in these scenarios
3. What information would narrow the gap, and when could you get it?
Before finalizing the Sprint 2 appendix, we could clarify the camera-service interface, define what “misuse” means, review the functional counts, and obtain team estimates for the integration tasks. Measured velocity would require completed implementation sprints and is unavailable for this planning-only activity.

Our explanation of the gap (one paragraph)

The Function Point method gives a middle estimate of 490 hours, while the Story Point method gives 552 hours. The difference reflects how functional counts and relative story sizes represent the work, together with the assumed productivity rates. Camera integration and the manager dashboard are particularly uncertain because their detailed requirements are not yet defined. The overlapping ranges support using a provisional planning range, but the shared assumptions mean that agreement does not validate either estimate.

Step 6. Present the estimate (Part 2, Slide 33)

This is the table your sponsor reads. It gives a range instead of a single number, shows what the range depends on, and is dated because estimates change as you learn more. It becomes the Estimation Appendix on your Sprint 2 portal page.

How to fill it in

1. Range: the lowest and highest hours across both methods.
2. Expected value: set O to your low end and P to your high end. Choose M, your most likely value, somewhere between them and explain why. Then calculate $(O + 4M + P) \div 6$.
3. Cost: multiply hours by a loaded hourly rate. State the rate and where it came from, such as published wage data plus an overhead percentage you choose.
4. Calendar time: divide hours by (team size $\times$ productive hours per person per week). State the productive hours you assume.

Field	Our estimate
Slice estimated	Parking Platform: all six stories in Step 1, including directions and filtering
Range in hours (low to high)	340–680 team hours, including integration and coordination
Expected hours: $(O + 4M + P) \div 6$, and why we chose M	O = 340/M = 552/ P = 680. $(340 + 4 \times 552 + 680) \div 6 = 538$ hours We provisionally use the Story Point middle estimate because it directly reflects the six sized stories. It is also the more cautious of the two middle estimates. This is a planning judgment, not a measured result.
Cost range and loaded rate used	Assume \$60 per person-hour, including labor and overhead, solely for this classroom estimate. $340 \times \$60$ to $680 \times \$60 = 20,400 - 40,800$. Expected labor cost: $538 \times \$60 = \$32,280$.
Calendar estimate in weeks, and hours per week assumed	Assume 4 people $\times$ 20 available project hours/week = 80 team hours/week. Range: $340 \div 80$ to $680 \div 80 = 4.25-8.5$ weeks. Expected: $538 \div 80 = 6.73$ weeks.
Methods used	Function Points and Story Points
Key assumptions	50 FP / 6–12 hours/FP - 32 story points; velocity of 8–16 points per two-week sprint; four people; 20 hours/person/week; 40 additional integration and coordination hours; an accessible camera service; a responsive website rather than separate native apps.
Team size assumed	4 people
What would change this estimate	Custom computer-vision development, unavailable camera data, more complex misuse detection, added stories, lower team availability, or revised productivity assumptions. Hardware and separately billed external services would require additional cost estimates.
Date of this estimate	September 29, 2026

Example, “Buy a semester permit” slice: range 354 to 708 hours. With $M = 520$, expected hours = $(354 + 2,080 + 708) \div 6 \approx 524$. At an assumed loaded rate of \$60 per hour, cost is about \$21,000 to \$42,500.

With 5 people at an assumed 30 productive hours a week (150 team hours), the expected 524 hours is about 3.5 weeks.

Going further (optional)
Once your slice estimate is solid, you can use it to make a rough estimate of the whole project. If your slice holds about one fifth of your backlog points, the whole project is roughly five times the slice. Treat that number as a first rough guess, and repeat this activity on more slices as you learn more.

Before you submit

1. Your slice is named, with its stories, screens, data, and boundary written down.
2. Both estimates are for the same slice and in hours, with the conversions shown.
3. Every assumed number is labeled as an assumption.
4. Every sourced number has a source you can link.
5. The team-size check is complete, and any No has a fix or explanation.
6. The final estimate is a dated range, not a single number.
7. The individual estimation memo is your own work, written without AI.